Mathematical and Computational Methods

Exercise Sheet 1: Foundations: numbers, sets and functions

Tutorial route (▸, in this order, about 80 minutes): 1.3(b,d), 2.1, 2.3(b,d), 2.4(b), 2.6(a,b), 2.8(a–c), 2.5. Physics groups: 2.9 in place of 2.8. Everything else is for home study.

Levels: ⋆ drill, ⋆⋆ core, ⋆⋆⋆ challenge. Give exact answers unless a decimal is asked for.

1 Warm-up drills

Exercise 1.1 *(⋆ drill — intervals & sets)*.

(a) Write $\{\, x \in \mathbb{R} \;:\; -2 \le x < 5 \,\}$ in interval notation.

(b) Write $(-\infty, 3]$ in set-builder notation.

(c) Express $\{\, x \in \mathbb{R} \;:\; x^2 \le 9 \,\}$ as a single interval.

(d) Compute $(-1, 4] \cap [2, 7)$.

(e) Compute $[0, 5] \setminus (2, 3)$.

Exercise 1.2 *(⋆ drill — Σ and Π notation)*. Evaluate

$$\text{(a)}\ \sum_{k=1}^{4} (3k-2), \qquad \text{(b)}\ \sum_{k=1}^{5} k^2, \qquad \text{(c)}\ \prod_{k=1}^{4} k, \qquad \text{(d)}\ \prod_{k=2}^{4} \left(1 - \tfrac{1}{k}\right).$$

Exercise 1.3 *(▸ ⋆ drill — inequalities)*. Solve for real x and give the answer as an interval (or union of intervals).

$$\text{(a)}\ 3 - 2x \le 7, \qquad \text{(b)}\ x^2 - x - 6 > 0, \qquad \text{(c)}\ |4x + 1| \le 7, \qquad \text{(d)}\ |x + 2| > 1.$$

Exercise 1.4 *(⋆ drill — maximal domain)*. State the maximal (natural) domain of each function.

$$\text{(a)}\ f(x) = \frac{1}{x-2}, \quad \text{(b)}\ g(x) = \sqrt{x-4}, \quad \text{(c)}\ h(x) = \ln(x+1), \quad \text{(d)}\ k(x) = \frac{1}{\sqrt{3-x}}.$$

Exercise 1.5 *(⋆ drill — composition)*. Let $f(x) = 2x + 1$, $g(x) = x^2$ and $k(x) = \dfrac{1}{x}$.

(a) Find $(f \circ g)(x)$, $(g \circ f)(x)$ and $(f \circ f)(x)$, each with its domain.

(b) Evaluate $(f \circ g)(3)$.

(c) Find $(k \circ k)(x)$ and its domain.

Exercise 1.6 *(⋆ drill — classify, then invert)*. For each map $f\colon \mathbb{R} \to \mathbb{R}$, state whether it is injective, surjective, both (bijective) or neither; for each bijective one, find f^{-1}.

$$\text{(a)}\ f(x) = 3x - 1, \quad \text{(b)}\ f(x) = x^2, \quad \text{(c)}\ f(x) = x^3, \quad \text{(d)}\ f(x) = \mathrm{e}^x.$$

(e) Find the inverse of $f(x) = \sqrt{x} + 1$ on $[0, \infty)$, with its domain.

Exercise 1.7 *(⋆ drill — even / odd / periodic)*. Classify each as even, odd, or neither; for (d) also state the period.

$$\text{(a) } x^4 - 3x^2, \quad \text{(b) } x^3 + x, \quad \text{(c) } x^2 + x, \quad \text{(d) } \cos(2x).$$

Exercise 1.8 *(⋆ drill — graph transformations)*. Describe in words how the graph of $y = f(x)$ moves to give (a) $y = f(x) + 3$, (b) $y = f(x - 2)$, (c) $y = -f(x)$, (d) $y = f(2x)$. Then, starting from $y = x^2$, identify the transformations giving $y = (x + 1)^2 - 4$ and state its vertex.

2 Tutorial problems

Exercise 2.1 *(► ⋆⋆ core — maximal domains and composites)*.

(a) Find the maximal domain of $r(x) = \dfrac{\sqrt{5 - x}}{x + 2}$, in set-builder and in interval notation.

(b) Let $f(x) = \sqrt{x}$ and $g(x) = 9 - x^2$. Find $(f \circ g)(x)$ and $(g \circ f)(x)$, each with its domain.

(c) Express $h(x) = (3x - 2)^4$ and $p(x) = \dfrac{1}{\sqrt{x^2 + 1}}$ each as a composition of simpler functions, naming the inner and the outer function.

Exercise 2.2 *(⋆⋆ core — inequalities, two ways)*. Solve

$$\text{(a) } \frac{2x + 1}{x - 3} \geq 0, \qquad \text{(b) } |x - 1| < |x + 3|\,.$$

Exercise 2.3 *(► ⋆⋆ core — x^2 and x^3: bijectivity and inverses)*.

(a) Show that $f(x) = x^2$, as a map $\mathbb{R} \to \mathbb{R}$, is neither injective nor surjective.

(b) Show that the same rule, as a map $[0, \infty) \to [0, \infty)$, is bijective, and write down its inverse.

(c) Explain why $f(x) = x^3$ is bijective on $\mathbb{R}$, and find f^{-1}.

(d) Find the inverse of $g(x) = (x - 1)^3 + 2$ and verify that $g\big(g^{-1}(x)\big) = x$.

Exercise 2.4 *(► ⋆⋆ core — graph transformations)*.

(a) Obtain $g(x) = 2(x - 3)^2 - 1$ from $y = x^2$ by a sequence of transformations; state the vertex and sketch both curves.

(b) Obtain $h(x) = 3 - \sqrt{x + 2}$ from $y = \sqrt{x}$. State the transformations used, the domain and the range of h, and whether h is increasing or decreasing.

Exercise 2.5 *(► ⋆⋆ core — even and odd parts* (maths)*)*.

(a) Prove that the product of two odd functions is even.

(b) Show that any function f on a domain symmetric about 0 can be written $f = E + O$ with $E(x) = \frac{1}{2}\big(f(x) + f(-x)\big)$ even and $O(x) = \frac{1}{2}\big(f(x) - f(-x)\big)$ odd. Apply this to $f(x) = \mathrm{e}^x$.

Exercise 2.6 *(► ⋆⋆ core — reading the piecewise ReLU)*. Recall $\mathrm{ReLU}(x) = \max(0, x)$.

(a) Write ReLU as a piecewise formula, and evaluate it at $x = -3,\ 0,\ 2$.

(b) Show that $\mathrm{ReLU}(x) = \frac{1}{2}\big(x + |x|\big)$.

(c) Hence express $|x|$ in terms of $\mathrm{ReLU}(x)$ and $\mathrm{ReLU}(-x)$.

Exercise 2.7 *(⋆⋆ core — trigonometric equations)*. Solve on $[0, 2\pi)$, giving exact answers in radians.

$$\text{(a) } 2\sin x - 1 = 0, \qquad \text{(b) } 2\cos^2 x - 1 = 0, \qquad \text{(c) } \sin x = \cos x.$$

Exercise 2.8 *(▶ ⋆⋆ core — the sigmoid as a probability (data science))*. The logistic (sigmoid) function is $\sigma(x) = \dfrac{1}{1+\mathrm{e}^{-x}}$.

(a) Compute $\sigma(0)$ and explain why $0 < \sigma(x) < 1$ for every real x.

(b) Verify the symmetry $\sigma(-x) = 1 - \sigma(x)$.

(c) In logistic regression a real-valued score s is turned into a class probability $\sigma(s)$. If $s = \ln 4$, find the probability. What score gives probability $\frac{1}{2}$?

(d) Sketch $y = \sigma(x)$ by hand, marking $\sigma(0)$, the two horizontal asymptotes, the direction of monotonicity and the centre of symmetry, each justified in one line (you may use (a) and (b)).

Exercise 2.9 *(⋆⋆ core — a potential barrier from steps (physics))*. The rectangular potential barrier of height $V_0 > 0$ and width $2a$ is $V(x) = V_0$ for $-a \le x < a$ and $V(x) = 0$ otherwise, as in the notes. (Which of the two endpoints carries the value V_0 is physically immaterial; this choice makes the step formula below exact.) Throughout, H is the Heaviside step, with the convention $H(0) = 1$ (so $H(x) = 1$ for $x \ge 0$ and $H(x) = 0$ for $x < 0$).

(a) What is the range of V? At which x-values does its graph jump?

(b) Show that $V(x) = V_0\big[H(x+a) - H(x-a)\big]$, and check the value at $x = 0$ and at $x = 2a$.

3 Further practice (home)

Exercise 3.1 *(⋆⋆ core — rational functions: graph, range, inequality)*.

(a) Starting from $y = \dfrac{1}{x}$, describe the graph of $y = \dfrac{1}{x-2} + 1$, giving its maximal domain and both asymptotes.

(b) Find the range of $f(x) = \dfrac{x^2}{x^2+1}$.

(c) Solve $\dfrac{x}{x+1} < 2$.

Exercise 3.2 *(⋆⋆ core — maximal domains, harder)*. State the maximal domain of each, in set-builder and in interval notation.

$$\text{(a) } f(x) = \ln(x^2 - 4), \quad \text{(b) } g(x) = \frac{\sqrt{x^2-1}}{x-3}, \quad \text{(c) } h(x) = \arcsin(x-1).$$

Exercise 3.3 *(⋆⋆ core — exponentials and logarithms: equations and graphs)*.

(a) Solve $3\,\mathrm{e}^{2x} = 12$.

(b) Solve $\log_2 x + \log_2(x-2) = 3$.

(c) Solve $\mathrm{e}^{2x} - 5\mathrm{e}^{x} + 6 = 0$.

(d) On one set of axes sketch $y = \mathrm{e}^x$, $y = \ln x$ and $y = x$, marking where each curve meets an axis and each asymptote, and explain why the two curves are reflections of each other in $y = x$. Hence sketch $y = \ln(x-1)$, stating its domain and asymptote.

Exercise 3.4 *(⋆⋆ core — tan, cot and the inverse trigonometric functions).*

(a) On separate axes, sketch $y = \tan x$ and $y = \cot x$ over $[-\pi, \pi]$. Mark every vertical asymptote and every x-intercept in that range, and state the period of each.

(b) arctan is the inverse of tan restricted to $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. State its domain and range, explain why $y = \pm\frac{\pi}{2}$ are horizontal asymptotes and why arctan is odd, evaluate $\arctan 0$ and $\arctan 1$, and sketch $y = \arctan x$.

(c) State the domain and range of arcsin, arccos and arccot. Evaluate $\arcsin\frac{1}{2}$, $\arccos\left(-\frac{1}{2}\right)$, $\operatorname{arccot} 0$ and $\operatorname{arccot} 1$, give the horizontal asymptotes of $y = \operatorname{arccot} x$, and sketch $y = \operatorname{arccot} x$.

Exercise 3.5 *(⋆⋆ core — hyperbolic functions from the definitions).* Work directly from $\sinh x = \frac{1}{2}(e^x - e^{-x})$ and $\cosh x = \frac{1}{2}(e^x + e^{-x})$.

(a) Evaluate $\sinh 0$, $\cosh 0$ and $\tanh 0$.

(b) Show that sinh is odd and cosh is even.

(c) Prove $\cosh^2 x - \sinh^2 x = 1$.

(d) Hence show that $\cosh x \geq 1$ for every real x, and find where equality holds.

Exercise 3.6 *(⋆⋆ core — the logit (data science)).* For the sigmoid $\sigma(x) = \dfrac{1}{1+e^{-x}}$ (range $(0, 1)$):

(a) Show that its inverse is the *logit* $\sigma^{-1}(p) = \ln\dfrac{p}{1-p}$, $0 < p < 1$.

(b) A classifier outputs probability $p = 0.8$. What score s produced it?

Exercise 3.7 *(⋆ ⋆ ⋆ challenge — the Gauss sum, and using Σ and Π).*

(a) Prove $\displaystyle\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$ by writing the sum forwards and backwards and adding, with no appeal to induction.

(b) Hence show that $\displaystyle\sum_{k=1}^{n}(3k+2) = \frac{3n^2+7n}{2}$, and evaluate it at $n = 10$.

(c) Evaluate $\displaystyle\prod_{k=1}^{n}\frac{k+1}{k}$ for general n.

Exercise 3.8 *(⋆ ⋆ ⋆ challenge — monotonic ⇒ injective, but not conversely).*

(a) Prove that a strictly increasing function is injective.

(b) Give an injective function on $[0, 2]$ that is *not* monotonic.

Exercise 3.9 *(⋆ ⋆ ⋆ challenge — irrationality of $\sqrt{3}$).* Prove that $\sqrt{3}$ is irrational, imitating the proof for $\sqrt{2}$ in the notes. (Every integer has the form $3m$, $3m+1$ or $3m+2$.)